

EDUCATION

Michigan State University, East Lansing, MI

*Bachelor of Science, **Electrical Engineering**, (Minor in Mathematics)*

Affiliations: IEEE MSU Student Branch (**Technical Director**)

GPA: 4.00

Expected May 2029

EXPERIENCE

MSU Solar Car Racing

Vehicle Controls and Sensing

East Lansing, MI

September 2025 – Present

- Developed and deployed STM32 embedded firmware in C++ for an on-vehicle controller, implementing real-time CAN and LIN communication for 5 vehicle signals and fault states, including turn signals, hazards, braking, and battery protection system (BPS) faults
- Engineered safety-critical vehicle control electronics from schematic capture and automotive component selection through impedance-controlled PCB layout in Altium Designer, validating MCUs, sensors, MOSFETs, and regulators against electrical and thermal operating constraints
- Integrated and validated vehicle hardware, firmware, and communication systems through fine-pitch PCB assembly and board-level debugging, isolating critical signal-routing faults with multimeter and oscilloscope to deliver a functional control system validated on the vehicle

RF-Photonics Lab, Michigan State University

Undergraduate Researcher

East Lansing, MI

June 2026 – Present

- Developed Python-based transmit and receive software using the USRP Hardware Driver (UHD) API and built 35+ GNU Radio flowgraphs for real-time digital signal processing, implementing FIR filtering, pulse shaping, symbol-timing recovery, and QPSK/16-QAM/256-QAM modulation on USRP B210 hardware
- Optimized hardware-software integration and real-time signal performance on the USRP B210, reducing square-wave rise/fall time from 135 ns to 45 ns by tuning sample rate, frequency, and gain while characterizing link quality using BER and EVM and debugging underflow and external environment induced phase-offset issues

Microtechnology Lab, Michigan State University

Undergraduate Researcher

East Lansing, MI

September 2025 – Present

- Developed bare-metal C/C++ firmware from scratch on a Teensy 4.1, implementing real-time ADC/DAC control loops and timing critical waveform generation, with Python scripting for streamlined system operations and MATLAB to analyze experimental data and generate detailed graphical outputs to optimize system performance
- Performed hardware bring-up and debug on a multichannel potentiostat, root-causing 6 layout and assembly faults with oscilloscope and multimeter to restore full 4-channel functionality

PROJECTS

- **ANC Engine:** Engineered a fixed-point FxLMS active noise cancellation engine that converges to -34 dB noise reduction in 50 ms certified against a self-built deterministic acoustic simulator and 12-case robustness suite
- **RISC-V CPU & SoC:** Designed a complete RV32I processor and SoC in Verilog that passes all 41 official RISC-V instruction tests and runs C compiled with stock GCC synthesized to a Gowin FPGA at 27 MHz
- **Automotive ECU Platform:** Built 4-node CAN-FD vehicle network on STM32 in C++ implementing embedded security via a secure bootloader supporting A/B image slots, ECDSA-P256 signature verification, and watchdog-based rollback, reflashing nodes over bus through ISO-TP (ISO 15765-2) transport and UDS (ISO 14229) diagnostic server
- **Game Boy Emulator:** Built a cycle-accurate Game Boy emulator in C++ passing 34/34 Blargg and 91/93 Mooneye timing tests, and rendered pixel-perfect dmg-acid2 output

SKILLS

- **Embedded & Firmware:** C, C++, Assembly, bare-metal firmware, ARM Cortex-M (STM32, Teensy), FreeRTOS
- **Automotive & Protocols:** CAN, CAN-FD, LIN, ISO-TP (ISO 15765-2), UDS (ISO 14229), I²C, SPI, Secure Boot
- **Hardware Design:** Altium Designer, KiCad, LTspice, schematic capture, PCB layout, fine-pitch soldering
- **Digital Design:** Verilog, RTL design, RISC-V ISA, FPGA synthesis, Icarus Verilog
- **Tools & Workflow:** Git, GitHub Actions, CMake, Ninja, GCC, Python, MATLAB
- **Signal Processing & SDR:** GNU Radio, UHD, USRP, HackRF, modulation and demodulation, BER characterization